Exercise

Question paper no:

Common program

1. Sp=(x+y)+x
2. V=(x+y)/(m*n)
3. X=(a+b)*(c+d)
4. V=(a*a)+(b*b)
5. S=(m+n)*p
6. T=(m*n)*p
7. V=(m*n)*(m/n)
8. P=(m/n)*(o*q)
9. T=(q*r)*z
10. V=(s*t)*(q*r)

If statement:

1. To find the given number is ever or odd.
2. To find the biggest of two numbers.
3. To find the smallest of two numbers.
4. To find the given number is positive or negative.
5. To find the given age is eligible for vote or not.
6. To find profit or loss and how far profit or loss.
7. To get 3 angles and find triangle formed or not.
8. To get a value and print absolute positive value.

If else statement:

1. To find the given number is ever or odd.
2. To find the biggest of two numbers.
3. To find the smallest of two numbers.
4. To find the given number is positive or negative.
5. To find the given age is eligible for vote or not.
6. To find profit or loss and how far profit or loss.
7. To get 3 angles and find triangle formed or not.
8. To get a value and print absolute positive value.

If else ladder:

1. To find biggest of three number.
2. To find smallest of three number.
3. To find the given number is positive or negative or neutral.
4. To find the given age is child, adult or senior citizen.
5. To do arithmetic functions
    - 1 (or) + = Addition
    - 2 (or) - = Subtraction

- 3 (or) * = Multiplication
- 4 (or) / = Division
- 5 (or) % = modules

6. To find the temperature in centigrade and display a suitable message according to the temperature
   - Temp less than 0 = freezing
   - Temp from 0 to 10 = very cold
   - Temp from 10 to 20 = cold
   - Temp from 20 to 30 = normal
   - Temp from 30 to 40 = hot
   - Temp more than 40 = very hot
7. To find equivalent day for the given number.
8. To find equivalent month for the given number.
9. To find the number of days in the month by identifying the month or the given number.
10. To find the student grade. (Note: all marks are greater than 35 to print grade else print fail).
11. To find the entered character is vowel or consonant.
12. Calculate sales commission

Sales	Commission
Up to 10000	0%
10000 to 20000	5%
20000 to 50000	10%
50000 to 100000	15%
100000 and above	20% + 500(bonus)

13. Calculate gross pay for employee

Basic salary	HRA	DA
Up to 10000	20%	80%
10000 to 20000	25%	90%
20000 and above	30%	95%

Gross pay = basic salary + HRA + DA

14. Calculate tax for purchasing

Purchasing amount	Tax per amount
Up to 10000	0%
10000 to 20000	10%
20000 and above	20%

15. Write a program to calculate and print the electricity bill of a given customer. The customer id, name and unit consumed by the user should take from the keyboard and display the total amount to pay to the customer.

Unit	Charge per unit
Up to 199	@1.20
200 to 400	@1.50
400 t0 600	@1.80
600 and above	@2.00

If bill exceed Rs.400 then a surcharge of 15% will be charged and minimum bill should be Rs.100

Nested if:

1. To find biggest of three number.
2. To find smallest of three number.
3. To find biggest of four number.
4. To find smallest of four number.
5. To find the student grade. (Note: all marks are greater than 35 to print grade else print fail).
6. Check whether given year is leap year or not.
7. Check whether the triangle is equilateral, scalene, or isosceles.
8. To check whether the triangle is valid or not if sides are given.

9. Calculate sales commission

Sales	Commission
Up to 10000	0%
10000 to 20000	5%
20000 to 50000	10%
50000 to 100000	15%
100000 and above	20% + 500(bonus)

10. Calculate gross pay for employee

Basic salary	HRA	DA
Up to 10000	20%	80%
10000 to 20000	25%	90%
20000 and above	30%	95%

Gross pay = basic salary + HRA + DA

11. Calculate tax for purchasing

Purchasing amount	Tax per amount
Up to 10000	0%
10000 to 20000	10%
20000 and above	20%

12. Write a program to calculate and print the electricity bill of a given customer. The customer id, name and unit consumed by the user should take from the keyboard and display the total amount to pay to the customer.

Unit	Charge per unit
Up to 199	@1.20
200 to 400	@1.50
400 to 600	@1.80
600 and above	@2.00

If bill exceed Rs.400 then a surcharge of 15% will be charged and minimum bill should be Rs.100

Switch case:

1. To find equivalent day for the given number.
2. To find equivalent month for the given number.
3. To find the number of days in the month by identifying the month or the given number.
4. Check whether an alphabet is vowel or consonant.
5. Write a program to calculate function (add, sub, div, mod).
6. Write a program to print entered number in word (0-9).
7. Write a program to input sum, rate, time and type of Interest ('S' for Simple Interest and 'C' for Compound Interest). Calculate and display the sum and the interest earned.

Note: Display the menu. Accept the user's choice

The Simple Interest (SI) and Compound Interest (CI) of a sum (P) for a given time (T) and rate (R) can be calculated as:

(a) SI = (p * r * t) / 100
(b) CI = P * ((1 + (R / 100)) T - 1)

8. Write a program to calculate Celsius, Fahrenheit, Kelvin.

Fahrenheit to Celsius	$c = (F - 32) \times \frac{5}{9}$
Celsius to Fahrenheit	$F = c \times \frac{9}{5} + 32$
Celsius to kelvin	$k = c + 273.15$
Kelvin to Fahrenheit	$F = (k - 273.15) \times \frac{9}{5} + 32$
Fahrenheit to kelvin	$k = (f - 32) \times \frac{5}{9} + 273.15$
Kelvin to Celsius	$c = k - 273.15$

LOOPS (while, do-while, for):

1. Print Natural number till n.
2. Print Even number till n.
3. Print Odd number till n.
4. Print number from -10 to -1.
5. Print natural number in reverse.
6. Print Even number in reverse.
7. Print odd number in reverse.
8. Print Sum of natural number till n
9. Print sum of odd number till n.
10. Print sum of even number till n.
11. $1^2+2^2+3^2+\ldots+n^2$
12. $1^2+3^2+5^2+\ldots+n^2$
13. $2^2+4^2+6^2+\ldots+n^2$
14. Print sum of cube number till n.
15. $1/1^2+1/2^2+1/3^2+\ldots+1/n^2$

16. $1/1^2+1/3^2+1/5^2+\ldots+1/n^2$
17. $1/2^2+1/4^2+1/6^2+\ldots+1/n^2$
18. Find Factorial value for n.
19. 1! + 2! + 3! +……+n!
20. 1! + 3! + 5! +……+n!
21. 2! + 4! + 6! +……+n!
22. 1/1! + 1/2! + 1/3! +……+1/n!
23. 1/1! + 1/3! + 1/5! +……+1/n!
24. 1/2! + 1/4! + 1/6! +……+1/n!
25. Print current and previous Odd number till n.
26. Print current and previous even number till n.
27. N^{th} Multiplication table till m.
28. Reverse s number.
29. Print each character in string.
30. Reverse a string.
31. Check whether giver number is palindrome or not.
32. Check whether giver number is Armstrong or not.
33. Check whether given number is Adam or not.
34. Check whether given number is Perfect number or Not.
35. Convert given number from decimal to binary.
36. Convert given number from decimal to octal.
37. Convert given number from decimal to hexadecimal.
38. Convert given number from binary to decimal.
39. Convert given number from binary to octal.
40. Convert given number from binary to hexadecimal.
41. Convert given number from octal to decimal.
42. Convert given number from octal to binary.
43. Convert given number from octal to hexadecimal.
44. Convert given number from hexadecimal to decimal.
45. Convert given number from hexadecimal to binary.
46. Convert given number from hexadecimal to octal.
47. Check whether given number is Prime number or Not.
48. Fibonacci series.(0,1,1,2,3,5,8,13…..n)
49. Pell series .$\mathbf{p_2=p_0+2p_1,\ p_0=0,\ p_1=1.}$ (0, 1, 2, 5, 12, 29….n).
50. Convert time,
    i. Hour to sec.
    ii. Min to hour
    iii. Sec to min.
51. Count number of digits of a number.
52. Calculate sum of digits of a number.
53. Calculate product of digits of a number.
54. Swapping first and last digit of a number.
55. Find positive, negative, and zero for n^{th} value.
56. Find biggest number for n time's value.

57. Find smallest of number for n times value.
58. Sum of numbers for n times value.
59. Sum of odd number for n times value.
60. Sum of even number for n times value.
61. Find sum and average n time's number.
62. Find highest common factor (HCF) for 2 numbers.
63. Find least common multiple (LCM) for 2 numbers.
64. Check whether given number is prime Adam or not.
65. Check whether given number Automorphic or not
66. Enter a=0, b=0, c=0 and get n value to print the following pattern.

```
000
001
002
003
.
.
.
010
011
.
.
.
nnn
```

Nested loop:

1. Print palindrome numbers from 100 to 999
2. Print prime numbers from 1 to 100.
3. Print Armstrong numbers from 100 to n.
4. Print Adam numbers from 100 to n.
5. Print Adam number from 1 to 100.
6. Automorphic numbers from 1 to 100.
7. Calculate student grade for each subject.
8. Find whether the given number is strong number.
9. Multiplication table chart.

Patterns:

1)
```
1
12
123
1234
12345
```

2)
```
5
54
543
5432
54321
```

3)
```
    1
   21
  321
 4321
54321
```

4)
```
1
23
456
78910
```

5)
```
    1
   21
  321
 4321
54321
```

6)
```
54321
4321
321
21
1
```

7)
```
10101
01010
10101
01010
```

8)
```
1
10
101
1010
10101
```

9)
```
1
12
123
1234
12345
1234
123
12
1
```

10)
```
1
22
333
4444
55555
```

11)
```
*
* *
* * *
* * * *
* * * * *
```

12)

```
12345
1234
123
12
1
```

13)

```
54321
5432
543
54
5
```

14)

```
    1
   12
  123
 1234
12345
```

15)

```
    *
   * *
  *   *
 *     *
*       *
 *     *
  *   *
   * *
    *
```

16)

```
12345
2345
345
45
5
```

17)

```
5
45
345
2345
12345
```

18)

```
1
2 6
3 7 10
4 8 11 13
5 9 12 14 15
```

19)

```
   *
  * *
 *   *
*     *
*********
```

20)

```
     1
    121
   12321
  1234321
 123454321
  1234321
   12321
    121
     1
```

21)

```
*
**
***
****
*****
****
***
**
*
```

22)

```
*********
*       *
 *     *
  *   *
   *
```

23)

```
A
AB
ABC
ABCD
ABCDE
```

24)

```
*****
*****
*****
*****
*****
```

25)

```
    *
   ***
  *****
 *******
*********
 *******
  *****
   ***
    *
```

26)

```
    A
   B B
  C   C
 D     D
E       E
F       F
E       E
 D     D
  C   C
   B B
    A
```

27)

```
*********
*       *
 *     *
  *   *
   * *
    *
   * *
  *   *
 *     *
*********
```

28)

```
*****
*   *
*   *
*   *
*****
```

29)

```
AB
ABC
ABCD
ABCDE
ABCDEF
```

30)

```
*
**
***
****
*****
******
```

31)

```
A
BB
CCC
DDDD
EEEEE
```

32)

```
ABCDE
ABCD
ABC
AB
A
```

33)

```
   *
  * *
 *   *
*******
```

34)

```
******
*****
****
***
**
*
```

35)

```
*
* *
*  *
*   *
*    *
******
```

36)

```
     *
    * *
   *   *
  *     *
 *       *
* * * * * *
```

37)

```
1
1 2
1   3
1     4
1 2 3 4 5
```

38)

```
1 2 3 4 5
2       5
3   5
4 5
5
```

39)

```
      1

    1 2 1

  1 2 3 2 1

1 2 3 4 3 2 1

1 2 3 4 5 4 3 2 1
```

40)

```
      *
     * *
    * * *
   * * * *
  * * * * *
 * * * * * *
```

41)

```
1
1 2
1 2 3
1 2 3 4
1 2 3 4 5
```

42)

```
      1
    2 3 2
  3 4 5 4 3
4 5 6 7 6 5 4
5 6 7 8 9 8 7 6 5
```

43)

```
      1

    1 2 1

  1 2 3 2 1

1 2 3 4 3 2 1

1 2 3 4 5 4 3 2 1
```

44)

```
********1********

*******2*2*******

******3*3*3******

*****4*4*4*4*****

****5*5*5*5*5****
```

45)

```
* * * * * *
 * * * * *
  * * * *
   * * *
    * *
     *
```

46)

```
1 2 3 4 5
1 2 3 4
1 2 3
1 2
1
```

47)

```
      1
    1  2
  1    3
1      4
1 2 3 4 5
```

48)

```
A

A B A

A B C B A

A B C D C B A

A B C D E D C B A
```

49)

```
***** *****
****   ****
***     ***
*         *
*         *
**       **
***     ***
****   ****
***** *****
```

50)

```
*                   *
* *               * *
* * *           * * *
* * * *       * * * *
* * * * *   * * * * *
* * * * * * * * * * *
* * * * * * * * * * *
* * * * *   * * * * *
* * *           * * *
* *               * *
*                   *
```

51)

```
    ****

   ****

  ****

 ****
```

52)

```
3
44
555
6666
555
44
3
```

53)

```
*****

*****

*****
```

54)

```
 * * * *

*     *

*     *

* * * *
```

55)

```
1

2*2

3*3*3

4*4*4*4

4*4*4*4

3*3*3

2*2

1
```

56)

```
* *  * * *

*         *

* * *  * *
```

57)

```
1
1 1
1 2 1
1 3 3 1
1 4 6 4 1
1 5 10 10 5 1
1 6 15 20 15 6 1
```

58)

```
1
2*3
4*5*6
7*8*9*10
7*8*9*10
4*5*6
2*3
1
```

59)

```
        *
      * * *
    *   *   *
  *     *     *
* * * * * * * * * *
  *     *     *
    *   *   *
      * * *
        *
```

60)

```
        *
      *
    *
  *
**********
  *
    *
      *
        *
```

ARRAYS

1) Find and print the 3rd and 5th element in the array,

2)Find and print the first and last element in the array

3) Change the 2nd number in the array to 50.

4)Let a={10,20,30,40,50} in this array swap the first and last element, like a={50,20,30,40,10)

3) Write a Java program to test if an array contains a specific value,

6) Write a Java program to find the index of an array element.

7) Write a Java program to remove a specific element from an array

8) Find the total of all the elements in the array and print it

9) Write a program for addition of two single dimensional arrays.

10) Write a Java program to reverse an array of integer values.

11)Write a program for arranging the numbers in the array in Ascending order,

12) Write a program for arranging the number in the array in Descending order.

13) Find the largest number in the array.

14) Find the smallest number in the array.

15) Find the 2nd largest number in the array while in ascending order.

16) Find the 2nd smallest number in the array while in ascending order

17) Find the 2nd largest number in the array while in descending order.

18) Find the 2nd smallest number in the array while in descending order.

19) Find the duplicate elements in the array.

20) Write a Java program to find the duplicate values of an array of string values.

21) Write a Java program to find the common elements between two arrays of integers.

22) Write a Java program to find the common elements between two arrays (string values).

23) Write a Java program to remove duplicate elements from an array.

24) Write a Java program to test the equality of two arrays.

25) Write a Java program to move all 0's to the end of an array. Maintain the relative order of the other

(non-zero) array elements.

26) Write a Java program to find the number of even and odd integers in a given array of integer.

2/) Write a Java program to compute the average value of an array of integers except the largest and

smallest values.

28) Write a Java program to check if an array of integers without 0 and -1.

29) Write a Java program to print all the LEADERS in the array.

Note: An element is leader if it is greater than all the elements to its right side.

30) Write a Java program to separate 0s on left side and 1s on right side of an array of 0s and 1s in

random order.

Multi-dimensional array:

1) Write a program for Transpose matrix.

Write a program for Matrix multiplication.

3) Find the given Matrix is whether Identity Matrix or not.

4) Write a program for Symmetric matrix.

5) Write a program to print the Student mark list in the table format using matrix.